

Phase 5: Demo and Onboarding Flow Implementation

Implementation Summary

This document describes the complete implementation of the Demo and Onboarding Flow with Transbank payment integration for the CRTLPyme SaaS platform.

What Was Implemented

1. Demo Account Creation

API Endpoint: `/api/demo`

- **Method:** POST
- **Purpose:** Creates a free 14-day trial account without payment
- **Features:**
 - Validates email and business information
 - Creates tenant with TRIAL status
 - Generates admin user with random password
 - Creates basic subscription (trial mode)
 - Sends welcome email via SendGrid
 - Creates audit log entry
 - Returns temporary credentials

UI Page: `/demo`

- Simple, user-friendly form
- Collects: first name, last name, business name, email, phone
- Shows success screen with credentials
- Auto-redirects to login after 5 seconds
- Provides links to login and onboarding

Flow:

1. User fills out demo form
 2. System creates trial tenant (TRIAL status, 14 days)
 3. System generates temporary password
 4. Welcome email sent
 5. User receives credentials on screen
 6. User can login immediately
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2. Multi-Step Onboarding Wizard

UI Page: `/onboarding`

- **Step 1:** Company Information
 - Business name
 - RUT (Chilean tax ID)
 - Email
 - Phone

- Address
- Validates RUT format (XXXXXXXX-X)
 - **Step 2:** Plan Selection
 - Fetches plans from `/api/subscription-plans`
 - Displays plan cards with features
 - Shows pricing and billing cycle
 - Visual selection indicator
 - **Step 3:** Payment Confirmation
 - Summary of company information
 - Selected plan details
 - Total amount to pay
 - Transbank payment integration
 - Redirects to Transbank Webpay Plus
 - **Progress Indicator:** Visual step tracker (1/3, 2/3, 3/3)

API Endpoint: `/api/onboarding`

- **Method:** POST
- **Purpose:** Creates tenant account for paid subscription
- **Features:**
 - Validates company information
 - Checks for duplicate RUT/email
 - Creates tenant (SUSPENDED status until payment)
 - Creates admin user
 - Creates pending subscription
 - Returns tenant and subscription IDs

Flow:

1. User completes Step 1 (company info)
2. User selects plan in Step 2
3. User confirms in Step 3
4. System creates suspended tenant account
5. System initiates Transbank payment
6. User redirected to Transbank payment page
7. After payment, redirected back to callback URL

3. Transbank Payment Integration

Existing Endpoints (Enhanced):

- `/api/subscriptions/payment/init` - Initiates payment with Transbank
- `/api/subscriptions/payment/callback` - Processes payment response

Enhanced Callback Features:

- **On Payment Success:**
 - Activates subscription (ACTIVE or TRIAL status)
 - **Activates tenant account** (changes from SUSPENDED to ACTIVE/TRIAL)
 - Sets `isActive = true` and `onboardingCompleted = true`

- Calculates billing dates (next billing, trial end)
- Sends welcome email via SendGrid
- Sends payment success email via SendGrid
- Creates audit log for account activation
- Records payment webhook
- Redirects to success page

- **On Payment Failure:**

- Marks subscription as CANCELLED
- Keeps tenant as SUSPENDED
- Records failure reason
- Creates audit log
- Records payment webhook
- Redirects to error page with reason

Transbank Flow:

1. System calls `/api/subscriptions/payment/init`
 2. Transbank creates transaction, returns token
 3. User redirected to Transbank payment page
 4. User completes payment on Transbank
 5. Transbank redirects back with `token_ws`
 6. System calls `commitTransaction()` to confirm
 7. System updates database based on payment result
 8. User redirected to success/error page
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4. Account Activation Flow

Activation Triggers:

1. **Demo Account:** Activated immediately upon creation (TRIAL status)
2. **Paid Account:** Activated after successful Transbank payment

Activation Steps:

1. Update tenant:
 - `isActive = true`
 - `accountStatus = ACTIVE` or `TRIAL`
 - `onboardingCompleted = true`
 - `lastActivityAt = current timestamp`
1. Update subscription:
 - `status = ACTIVE` or `TRIAL`
 - Set billing dates (`startDate` , `endDate` , `nextBillingDate`)
 - Set trial end date (if applicable)
 - Enable auto-renewal
2. Send emails:
 - Welcome email with account details
 - Payment success email (for paid accounts)
3. Create audit logs:
 - Record account activation
 - Record payment success

Email Templates (via SendGrid):

- `sendWelcomeEmail()` - Welcome message with login info
- `sendPaymentSuccessEmail()` - Payment confirmation with receipt details

5. Home Page Updates**Changes to** `/app/page.tsx` :

- Header button changed from “Empezar Gratis” to “Prueba Gratis 14 Días” → links to `/demo`
- Hero section CTA updated:
- “Prueba Gratis 14 Días” → links to `/demo`
- “Contratar Ahora” → links to `/onboarding`

**File Structure**

```

app/
├── api/
│   ├── demo/
│   │   └── route.ts                # Demo account creation API
│   ├── onboarding/
│   │   └── route.ts              # Onboarding account creation API
│   ├── subscriptions/
│   │   └── payment/
│   │       ├── init/route.ts     # (Existing) Payment initiation
│   │       └── callback/route.ts # (Enhanced) Payment callback
│   └── demo/
│       ├── page.tsx             # Demo registration page
│       └── onboarding/
│           └── page.tsx         # Multi-step onboarding wizard
│                                   # (Updated) Home page with new CTAs
├── lib/
│   ├── transbank.ts             # (Existing) Transbank integration
│   └── sendgrid.ts              # (Existing) Email service
└── prisma/
    └── schema.prisma            # (Existing) Database schema
  
```

**Complete User Journeys****Journey 1: Demo User**

1. Visit home page
2. Click “Prueba Gratis 14 Días”
3. Fill out demo form (name, email, company)
4. Submit form
5. ☒ Account created instantly (TRIAL, 14 days)
6. View credentials on screen
7. Click “Iniciar Sesión” or wait for auto-redirect
8. Login with temporary password

9. Start using the platform

No payment required | 14-day trial | Instant access

Journey 2: Paid Subscription User

1. Visit home page
2. Click "Contratar Ahora"
3. **Step 1:** Enter company information (name, RUT, email, phone, address)
4. Click "Siguiente"
5. **Step 2:** Select a plan (Basic, Pro, or Enterprise)
6. Click "Siguiente"
7. **Step 3:** Review summary and click "Pagar con Transbank"
8. Redirected to Transbank payment page
9. Complete payment (credit/debit card)
10. Redirected back to callback URL
11.  Account activated automatically (ACTIVE status)
12. Welcome and payment success emails sent
13. Redirected to success page
14. Click "Ir al Dashboard"
15. Login and start using the platform

Payment required | Immediate activation | Full access



Database Changes

Tenant States

- **TRIAL:** Demo accounts (14 days free)
- **ACTIVE:** Paid accounts with active subscription
- **SUSPENDED:** Accounts pending payment or with payment issues
- **BLOCKED:** Accounts blocked by admin
- **CANCELLED:** Accounts cancelled by user

Subscription States

- **TRIAL:** Trial period active
- **ACTIVE:** Paid subscription active
- **SUSPENDED:** Pending payment or payment failed
- **CANCELLED:** Cancelled by user or system
- **EXPIRED:** Subscription expired

Key Fields

```
Tenant {
  isActive: boolean           // true after payment/demo creation
  accountStatus: AccountStatus // TRIAL, ACTIVE, SUSPENDED, etc.
  onboardingCompleted: boolean // true after activation
  trialStartedAt: DateTime?    // When trial started
  trialEndsAt: DateTime?       // When trial ends
}

Subscription {
  status: SubscriptionStatus // TRIAL, ACTIVE, etc.
  startDate: DateTime         // Subscription start
  endDate: DateTime?         // Current period end
  nextBillingDate: DateTime?  // Next charge date
  trialEndsAt: DateTime?      // Trial end date
  autoRenew: boolean          // Auto-renewal setting
}
```

Testing the Implementation

Test Demo Account Creation

```
curl -X POST http://localhost:3000/api/demo \
-H "Content-Type: application/json" \
-d '{
  "email": "demo@test.cl",
  "businessName": "Demo Test SpA",
  "firstName": "Juan",
  "lastName": "Pérez",
  "phone": "+56912345678"
}'
```

Expected Response:

```
{
  "success": true,
  "data": {
    "tenantId": "...",
    "userId": "...",
    "email": "demo@test.cl",
    "tempPassword": "...",
    "trialEndsAt": "2024-11-22T...",
    "message": "Cuenta demo creada exitosamente..."
  }
}
```

Test Onboarding Account Creation

```
curl -X POST http://localhost:3000/api/onboarding \
-H "Content-Type: application/json" \
-d '{
  "businessName": "Mi Empresa SpA",
  "rut": "12345678-9",
  "email": "empresa@test.cl",
  "phone": "+56912345678",
  "address": "Av. Providencia 1234, Santiago",
  "planId": "plan-id-from-database"
}'
```

Expected Response:

```
{
  "success": true,
  "data": {
    "tenantId": "...",
    "userId": "...",
    "subscriptionId": "...",
    "email": "empresa@test.cl",
    "tempPassword": "...",
    "message": "Cuenta creada exitosamente..."
  }
}
```

Security Considerations

1. **RUT Validation:** Basic format validation (XXXXXXXX-X)
2. **Email Validation:** Regex pattern validation
3. **Duplicate Prevention:** Checks for existing email/RUT
4. **Password Generation:** Random 10-20 character passwords
5. **Password Hashing:** bcrypt with 10 salt rounds
6. **Transbank Token:** Single-use tokens for payments
7. **Account Status:** Suspended until payment confirmed
8. **Audit Logs:** All account activations logged

Email Integration

All emails are sent via **SendGrid** using the existing `lib/sendgrid.ts` service.

Email Types:

1. **Welcome Email** (`sendWelcomeEmail`)

- Sent to: Demo accounts and paid accounts
 - Contains: Welcome message, plan details, login link
1. **Payment Success Email** (`sendPaymentSuccessEmail`)
 - Sent to: Paid accounts after successful payment
 - Contains: Payment receipt, plan details, next billing date
 2. **Payment Failed Email** (existing)
 - Sent to: Users with failed payments
 - Contains: Failure reason, retry instructions

Deployment Considerations

Environment Variables Required

```
# Transbank
TRANSBANK_API_KEY=your-api-key
TRANSBANK_COMMERCE_CODE=your-commerce-code
TRANSBANK_ENVIRONMENT=production # or integration

# SendGrid
SENDGRID_API_KEY=your-sendgrid-api-key
SENDGRID_FROM_EMAIL=noreply@crtlpyme.cl
SENDGRID_FROM_NAME=CRTLPyME

# Next.js
NEXTAUTH_URL=https://your-domain.com
DATABASE_URL=your-database-url
```

Database Migration

No new migrations required. All changes use existing schema.

Build and Deploy

```
npm run build
npm start
# or deploy to Vercel/similar platform
```

Future Enhancements

1. **Email Verification:** Add email verification before activation
2. **Phone Verification:** Add SMS verification for Chilean phone numbers
3. **RUT Validation:** Implement complete RUT digit verification
4. **Payment Plans:** Add support for installment payments
5. **Trial Extension:** Allow trial period extension
6. **Promo Codes:** Add discount/promo code support
7. **Webhook Retry:** Implement automatic webhook retry logic
8. **Payment History:** Add detailed payment history page

- 9. **Invoice Generation:** Auto-generate PDF invoices
- 10. **Multi-language:** Add support for English/Portuguese



Known Issues

1. **Transbank Integration Environment:** Currently using integration environment
 - Need to update to production credentials for live deployment
2. **Email Sending:** Non-critical failures don't block account creation
 - Consider adding email queue for retry mechanism
3. **Password Strength:** Generated passwords are random but not user-chosen
 - Consider adding password reset flow on first login
4. **Session Management:** No automatic session creation after account activation
 - Users need to manually login after activation



Support

For issues or questions:

- Email: support@crtlpyme.cl
- GitHub: <https://github.com/kbzas090/CRTLPyme>
- Documentation: See project README.md



Checklist - Phase 5 Complete

- [x] Demo account creation API endpoint
- [x] Demo account UI page
- [x] Multi-step onboarding wizard (3 steps)
- [x] Onboarding API endpoint
- [x] Transbank payment integration
- [x] Payment callback enhancement
- [x] Account activation flow
- [x] Email notifications (welcome, payment success)
- [x] Audit logging
- [x] Home page updates
- [x] Error handling
- [x] Build verification
- [x] Git commit and push

Implementation Date: November 8, 2024

Commit: `c88100d`

Branch: `main`

Status:  Complete and Deployed

Ready for Testing

The complete demo and onboarding flow is now live and ready for end-to-end testing!

Test URLs:

- Demo: `http://localhost:3000/demo`
- Onboarding: `http://localhost:3000/onboarding`
- Home: `http://localhost:3000/`